

Grid Forming and Grid Following Control for Frequency and Voltage Stabilization of Hybrid Renewable Energy Systems in Microgrid

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Abstract—This study presents a grid-forming (GFM) inverter designed for a battery energy storage system (BESS) to maintain voltage and frequency stability within an AC microgrid. The proposed GFM inverter plays a critical role in autonomously regulating voltage and frequency when the utility grid experiences failures or disruptions. Unlike grid-following (GFL) inverters, which rely on an external grid reference, the GFM inverter takes control during grid outages, forming a stable local grid by dynamically adjusting its output. This ensures uninterrupted microgrid operation, enabling seamless transition between grid-connected and islanded modes. The battery storage system, managed through the GFM inverter, supports frequency stabilization by balancing power fluctuations, thereby enhancing microgrid reliability and resilience. The AC microgrid network consists of a wind-generating system (WGS), photovoltaic system (PVS), BESS, three-phase utility grid, AC loads, AC/DC converters, and DC/DC converters. Maximizing power extraction methodologies augment energy production from both WGS and PVS. A GFL inverter employing active and reactive power regulation is executed with a Phase Lock Loop (PLL) and vector control methodology to achieve synchronization with the utility grid. The GFM inverter utilizes droop control to maintain steady voltage and frequency for effective grid integration. The system is validated using simulations under different scenarios, including the sudden implementation of WGS and grid failures. The rapid frequency response illustrates the efficacy of the proposed control approach. Simulation results provide a swift frequency response and maintain grid stability, validating its appropriateness for AC microgrids.

Keywords—Grid forming; Grid following; GFM; GFL; renewable energy resources; droop control; grid-connected.

I. INTRODUCTION

Renewable energy systems (RESs), including solar photovoltaic (PV) and wind generating systems (WGS), are progressively included in microgrids to improve energy sustainability and reliability [1][2][3]. These systems generally depend on power electronics converters to regulate the conversion and distribution of electrical energy inside the microgrid, as elaborated in [4]–[6]. The use of these converters presents considerable issues in control and administration owing to the intrinsically variable characteristics of renewable energy sources and the swift swings in load power demands within the microgrid. A principal problem is the necessity for control methods that can react promptly and precisely to these variations. The utility grid and load demand can fluctuate rapidly, often within milliseconds, necessitating that the control systems for these converters be highly dynamic and capable of functioning at fast sampling rates, sometimes

within only microseconds. This swift reaction is crucial for preserving system stability, guaranteeing effective power distribution, and averting disruptions inside the microgrid. Further, efficient control systems for these power electronics converters have to include system restrictions, including voltage and current limits as well as safety margins, to guarantee safe and reliable operation. Neglecting these limits may result in operational problems, including possible damage to the converters and other microgrid components, hence reducing the system's reliability and durability.

Hence, GFL control is a widely adopted approach for integrating inverter-based renewable energy sources, such as PVS and WGS, into the power grid. This approach simplifies inverter design and enables effective power transfer to the grid by using the grid voltage and frequency as a reference for operation. GFL inverters provide active and reactive power by synchronizing with the grid's phase angle through the use of phase-locked loop (PLL) technology. Although this control method works well in synchronous generator-dominated stable grids, it has trouble maintaining stability and resilience in weak or low-inertia systems [7]. In addition, in modern grids with high renewable energy penetration, their dependency on a stable grid limits their ability to operate during disturbances or in standalone applications, such as microgrids. Moreover, they do not contribute to grid stability through inertia or frequency regulation, making them less suitable for grids with reduced synchronous generation [8]. Despite these limitations, GFL control remains a cost-effective and straightforward solution for renewable energy integration in traditional power systems.

Consequently, grid-forming (GFM) control, which can stabilize and smoothly incorporate RESs into electrical grids, has been developed to address the shortcomings of the GFL [9]. The GFM inverters function as a virtual synchronous generator by establishing voltage and frequency. Because of this ability, GFM control is crucial for preserving grid stability, particularly in systems where synchronous generators' traditional function is reduced due to a large penetration of renewable energy sources like wind and photovoltaics [10]. To guarantee steady operation under a range of grid situations, the main goal of GFM control is to emulate the dynamic behavior of traditional generators, including inertia and droop characteristics. GFM inverters can operate in both grid-connected and islanded modes, making them highly versatile for applications in microgrids, hybrid energy systems, and weak grid environments. As the transition toward renewable energy accelerates, GFM control is increasingly being recognized as a cornerstone technology for enabling the transformation of traditional grids into

decentralized, flexible, and sustainable energy systems. Although GFM control has many benefits for integrating RESs and stabilizing contemporary power networks, it also has drawbacks and difficulties because it depends on ESS to supply the power required to regulate frequency and voltage. This reliance may raise system expenses and create new battery management difficulties. Furthermore, grid-forming inverter control methods are more intricate and computationally intensive.

Therefore, this paper presents a coordinated control approach between GFL and grid-forming inverters for a hybrid photovoltaic/wind generation, BESS in a microgrid to attract the advantages of the two methods. The proposed integration aims to continue supplying power to the main load even though the utility grid is disconnected. In addition, maintaining the system stable under different conditions.

II. PROPOSED SYSTEM WITH MODELING

The microgrid system used in this investigation is shown in Fig. 1. A WGS, PVS, BESS, three-phase utility grid, AC loads, AC/DC converters, and DC/DC converters make up the microgrid system. Every part is outfitted with a unique control unit designed to govern how it operates. Among the tasks assigned to these control units are maximizing power generation, maintaining system stability, and adjusting to changes in load demands, grid outages, or environmental circumstances. In the following sections, the mathematical model of each component will be discussed.

A. PV mathematical model.

One important RES in the suggested microgrid system is the photovoltaic array. The two main elements that affect the PV output power and efficiency are solar radiation and operating temperature [11], [12] as described in

$$P_{PV} = f_{PV} P_{PVn} (1 + \beta_r (T_{PV} - T_{PV,STC})) \times (G_T / G_{T,STC}) \quad (1)$$

where T_{PV} denotes the actual temperature of the photovoltaic cell, while $T_{PV,STC}$ represents the temperature of the PV cell under standard test conditions. In addition, P_{PVn} represents the nominal power, while γ signifies the intensity coefficient for solar cell efficiency.

B. WG mathematical model.

In the adopted WGS, the permanent magnet synchronous generator (PMSG) is used because of its efficiency and reliability in wind energy applications [13]. A controlled rectifier is utilized to maximize the electrical output from the wind turbine (WT). This rectifier converts the alternating current generated by the PMSG into direct current (DC) power. The WT output power is affected by numerous critical parameters such as wind velocity, turbine blade radius, PMSG efficiency, and rectifier performance. The correlation between these variables and the power output is formally articulated in the following equations [14].

$$P_{WT} = 0.5 \rho C_p A V_\omega^3 \eta_{WT} \eta_G \quad (2)$$

$$C_p = 0.5176 \left(\frac{116}{\lambda_i} - 5 - 0.4\beta \right) e^{\frac{21}{\lambda_i}} + 0.0068\lambda_i \quad (3)$$

$$\lambda = \frac{\omega R}{V_\omega}, \quad \lambda_i^{-1} = \frac{1}{0.08\beta + \lambda} - \frac{0.035}{\beta^3 + 1} \quad (4)$$

where λ , R , and V_ω are the tip speed ratio, blade radius, and wind speed.

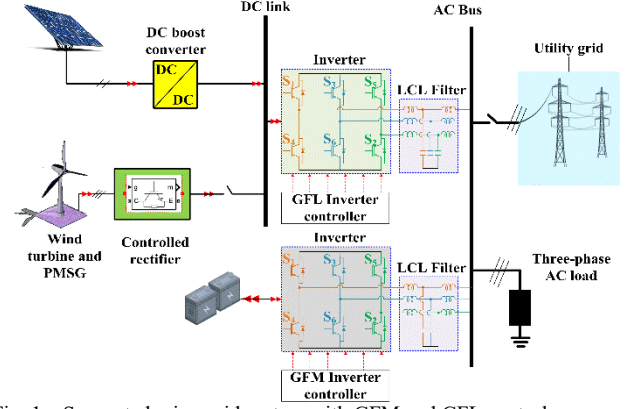


Fig. 1. Suggested microgrid system with GFM and GFL control.

C. Battery energy storage system (BESS)

The battery model employs a regulated voltage source alongside a fixed resistance, as illustrated in (5) and (6) [15].

$$E = E_0 - \left(P_v C_{BSS} \left/ \left(C_{BSS} - \int_0^t i_{BSS} dt \right) \right) + A \exp \left(-B \int_0^t i_{BSS} dt \right) \quad (5)$$

$$V_{BSS} = E - R_i I_{BSS} \quad (6)$$

where E denotes the no-load voltage, E_0 signifies the battery's constant voltage, P_v represents the polarization voltage, C_{BSS} indicates the battery capacity, $\int i_{BSS} dt$ reflects the actual battery charge. Meanwhile, the battery capacity is calculated using (7) [16].

$$C_{BSS} = E_{d/hr} / (V_{BSS} \times DoD) \quad (7)$$

where $E_{d/hr}$ represents the energy supplied to the load over one hour. DoD denotes the battery's depth of discharge.

III. CONTROL METHODOLOGY FOR THE SYSTEM COMPONENTS

A. Maximum power control for PVS

The maximum power point tracking (MPPT) is used to boost the overall PVS efficiency. The PVS employs the incremental conductance (IC) approach to extract the maximum power. The IC method's guiding concepts are explained in [17] and shown in Fig. 2. The mathematical representations are described by the following relations.

$$dP_{PV} / dV_{PV} = d(I_{PV} V_{PV}) / dV_{PV} = I_{PV} + V_{PV} (dI_{PV} / dV_{PV}) = 0 \quad (8)$$

$$(dI_{PV} / dV_{PV}) = -(I_{PV} / V_{PV}) \quad (9)$$

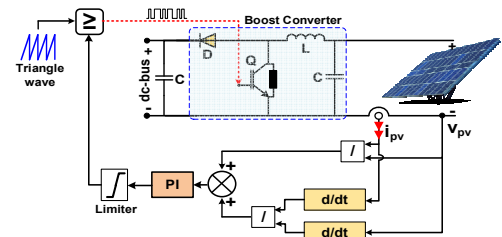


Fig. 2. MPPT of the PV system.

B. Maximum power control for WGS

To maximize the power generated from the WGS, the MPPT technique is used. In this system, a controlled rectifier is used instead of a switch-mode rectifier hence, dq -axis currents are utilized to achieve this goal. The maximum power is obtained for the analyzed wind turbine characteristics at a tip-speed ratio of 8.1, which corresponds to a wind power coefficient of 0.48 [18]. Hence, the maximum power extraction is accomplished with the reference current value on

the q -axis and to minimize losses, the d -axis current is rest to zero [19]. Using the ideal wind power, P_{opt} , the reference q -axis reference current, i_{qs}^* is provided by

$$P_{opt} = 0.5 \times C_{p-opt} \times \rho \times A \times V_w^3 \quad (10)$$

$$i_{qs}^* = (4P_{opt}) / (3p\omega\lambda_m) \quad (11)$$

Two Proportional-Integral (PI) controllers are employed to verify that the actual values accurately follow their reference points. Moreover, the compensation is utilized to achieve a total decoupling. Figure 3 illustrates the adopted MPPT control for the WGS.

C. Grid-following inverter control

As previously discussed, the GFL approach streamlines inverter design while enabling efficient power transfer to the grid by leveraging the grid's voltage and frequency as operational references. By synchronizing with the grid's phase angle through the PLL technology, the GFL approach effectively facilitates the delivery of both active and reactive power. Hence, the control is composed of three key components: PLL control, DC-bus voltage regulation, and grid current control. Each of these components plays a vital role in ensuring consistent, reliable, and grid-compliant operation. The detailed functionality and design of these controllers will be explored in the following subsections.

• Phase-locked loop

The primary goal of the PLL is to extract the grid voltage angle to properly align the d -axis voltage with the voltage vector in the voltage-oriented control (VOC) [20]. Also, the grid voltage on the q -axis is set to zero by the VOC principles. Because the PLL function maintains the grid voltage's d -axis with the voltage vector, the q -axis voltage is calculated by

$$v_{qg} = \hat{v}_{ph1} \sin(\omega_t - \omega) \quad (12)$$

where ω_{t0} and ω_t stand for the voltage vector and the d -axis's respective angles.

• DC-bus voltage control

By regulating the grid d -axis current, the DC-bus voltage is controlled. The input-output power equilibrium is used to illustrate this, presuming the system is lossless, as given by

$$\frac{3}{2}(v_{qg}i_{qg} + v_{dg}i_{dg}) = C.v_{dc} \cdot \frac{dv_{dc}}{dt} \quad (13)$$

Using the VOC concept, Eq. (13) is updated as follows

$$\frac{3}{2}v_{dg}i_{dg} = C.v_{dc} \cdot \frac{dv_{dc}}{dt} \quad (14)$$

By using linearization at the equilibrium point and removing the steady-state and higher-order terms, Eq. (14) is given by

$$\frac{3}{2}\hat{v}_{dg}I_{dg} + \frac{3}{2}V_{dg}\hat{i}_{dg} = C.V_{dc} \cdot \frac{dv_{dc}}{dt} \quad (15)$$

where $\hat{v}_{dg}$ and $\hat{i}_{dg}$ denote the perturbations in the d -axis voltage and current, respectively. Therefore, the value of the d -axis output current determines the DC-bus voltage. Hence, a PI controller is used to control the DC-bus voltage.

• Current-control loop

The reference value of the q -axis current is chosen to achieve unity, leading, or lagging power factor after the reference d -axis current is generated from the DC-bus voltage control loop. The inverter dq -axis output voltages can be expressed by [21]

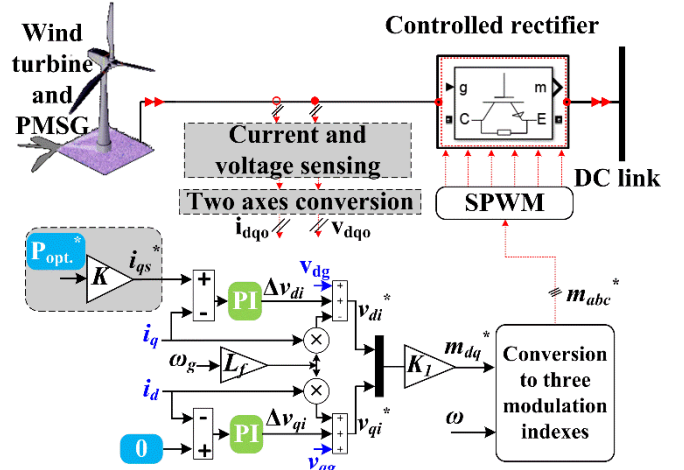


Fig. 3. MPPT control for the WGS.

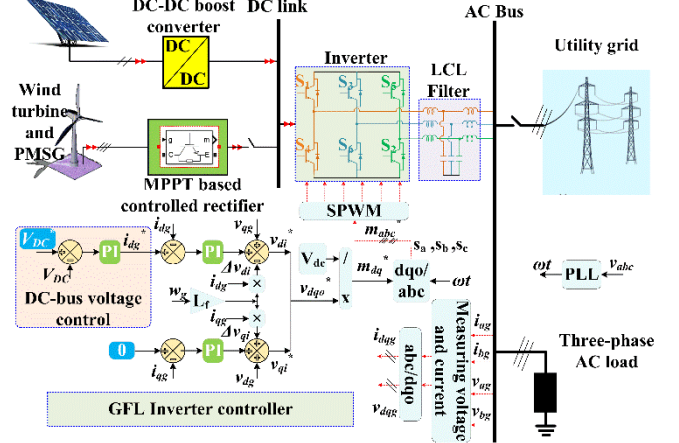


Fig. 4. Grid following inverter controller with the DC-link voltage control.

$$v_{di} = v_{dg} + R_i i_{di} + L_i p i_{di} + R_g i_{dg} + L_g p i_{dg} - \omega L_i i_{qi} \quad (16)$$

$$v_{qi} = v_{qg} + R_i i_{qi} + L_i p i_{qi} + R_g i_{qg} + L_g p i_{qg} + \omega L_i i_{di} \quad (17)$$

where R_i , R_g , L_i , and L_g are the filter resistances and inductances, and ω represents the grid angular frequency.

Equations (16) and (17) can be updated as in (18) and (19), and hence, two PI controllers can be used to generate the inverter's dq -axis output voltages.

$$v_{di}^* = \Delta v_{di} + v_{dg} - \omega L_i i_{qi} \quad (18)$$

$$v_{qi}^* = \Delta v_{qi} + v_{qg} + \omega L_i i_{di} \quad (19)$$

Figure 4 illustrates the grid following inverter controller with the DC-link voltage control block diagram.

D. Grid forming inverter control

GFM inverter, specifically designed for islanded operation, plays a crucial role in regulating voltage amplitude and frequency to establish a stable local AC grid. As a result, it can be configured as an ideal AC voltage source with low output impedance. While synchronization is a standard requirement for grid-connected inverters, GFM inverters demand a highly precise synchronization mechanism to operate effectively alongside other inverters. Typically, the phase reference is determined by integrating the defined angular frequency. The control strategy implemented in GFM inverters is developed to ensure both stable and efficient operation of the system. This approach facilitates the

maintenance of nominal voltage and frequency levels, in addition to regulating the active and reactive power distribution within the microgrid. When operating in grid-connected mode, the voltage and frequency are dictated by the main grid. However, in islanded mode, a dedicated control mechanism takes over to manage these parameters effectively [22]. The suggested control operates at multiple levels: the primary level is responsible for continuous real-time monitoring of the system's nominal voltage and addressing power quality issues [23]. The secondary level incorporates key control mechanisms that focus on system stability, voltage regulation, and frequency control [24]. Finally, the tertiary level consists of an outer voltage control loop and an inner current control loop.

Droop control is widely utilized in distributed generation networks to enable effective load sharing among parallel-connected inverters. This approach is particularly advantageous as it operates independently of communication links between inverters, ensuring decentralized control. Droop control serves as the predominant external control strategy implemented in nearly all experimentally validated microgrids [25]. The primary mechanism for droop control involves the regulation of both active and reactive power [26]. The standard droop characteristics are illustrated in Figs. 5, where the droop gains (K_p and K_q) are determined based on the specific characteristics of the system [27]. In the case of a predominantly inductive grid, such as a high-voltage network, the conventional droop control equations can be formulated as follows [28].

$$\omega = \omega_{ref} - K_p (P_{out} - P_{ref}) \quad (20)$$

$$V = V_{ref} - K_q (Q_{out} - Q_{ref}) \quad (21)$$

where V and ω represent the amplitude and angular frequency of the inverter's output voltage, respectively, while V_{ref} and ω_{ref} denote the rated output voltage and frequency of the inverter under no-load conditions. P_{out} and P_{ref} correspond to the feedback and reference values for the active power, whereas Q_{out} and Q_{ref} represent the feedback and the reference values for reactive power (refer to Fig. 7), respectively.

Furthermore, in addition to noise suppression, the active power droop mechanism is intended to emulate the inertia effect observed in synchronous machines [29]. The grid-forming inverter controller for BESS based on droop control is shown in Fig. 7. The selection process of the droop controller coefficients involves defining the permissible deviations in voltage and frequency corresponding to the inverter's output power range [30]. Beginning with the voltage droop coefficient, the maximum and minimum voltage levels are defined by

$$\begin{cases} V_{max} = (1 + \Delta v) V_n \\ V_{min} = (1 - \Delta v) V_n \end{cases} \quad (22)$$

where V_n is the nominal voltage. Since the inverters are positioned at a distance from the load, a voltage drop occurs along the transmission line, resulting in a lower voltage at the load. Hence, the droop coefficient is adjusted for the compensation.

$$K_q = \Delta v (V_{max} / Q_{rated}) \quad (23)$$

Similarly, the frequency droop coefficient can be obtained, where the system frequency fluctuates between defined maximum and minimum values. Because the virtual oscillator control is tuned to a resonant frequency of 50 Hz, the maximum frequency aligns with the nominal value. Consequently, the resulting frequency deviation and the droop coefficient are given by

$$\begin{cases} \omega_{max} = \omega_n \\ \omega_{min} = (1 - \Delta\omega) \omega_n \end{cases} \quad (24)$$

$$K_p = \Delta\omega (\omega_{max} / P_{rated}) \quad (25)$$

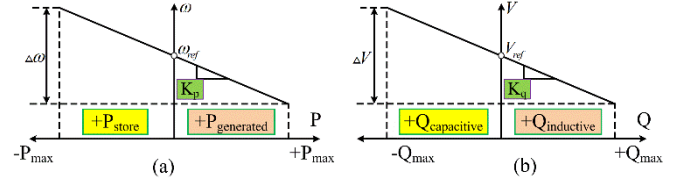


Fig. 5. Droop control strategies. (a) (P - ω) droop characteristics (b) (Q - V) droop characteristics.

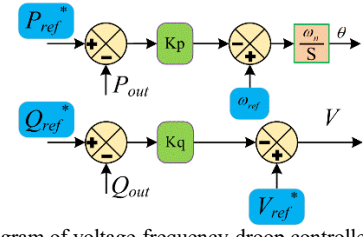


Fig. 6. Block diagram of voltage-frequency-droop controller.

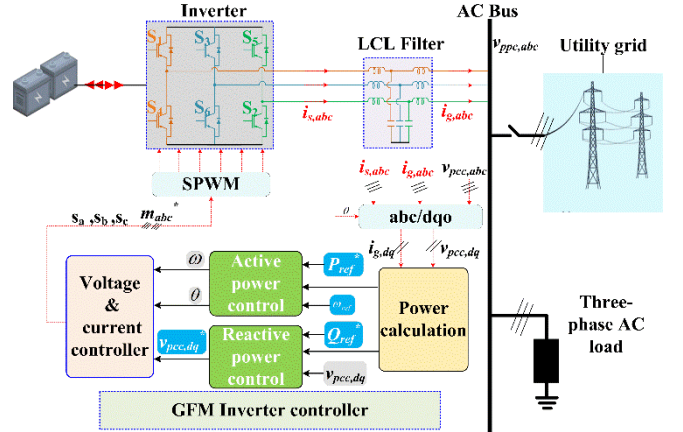


Fig. 7. Grid forming inverter controller for BESS based on droop control.

IV. SIMULATION RESULTS

This section presents the assessment of the proposed HMGES. The elements of the system were simulated in MATLAB/Simulink 2023 with a 20-kW wind generation system, a 30 kW photovoltaic system, and a 30 kW battery energy storage system. The technical study seeks to: (i) optimize power extraction from renewable sources, (ii) supply the generated power to the load, and (iii) keep the system stable even when there is an outage of the grid while ensuring a unity power factor and steady DC-bus voltage, irrespective of loading circumstances. This scenario aims to demonstrate that the grid-forming inverter provides swift frequency response and maintains grid stability, validating its appropriateness for contemporary AC microgrids. The overall performance of the GFM and the GFL inverter control and the system is tested under varying constant solar radiation, sudden application of constant wind speed, and

outage of AC grid. A steady wind speed of 8 m/s is sustained and the solar radiation of 1000 W/m² is adopted.

Figure 8 illustrates the power generation and consumption dynamics within the system, showcasing contributions from the photovoltaic system (PVS), wind generation system (WGS), battery energy storage system (BESS), utility grid, and load. It is evident that the maximum power is generated by the PVS and WGS. Initially, the PVS begins supplying power from the start, while the WGS is integrated into the system at $t = 2$ s. During the first 2 seconds, the total load demand of 30 kW is met by contributions from the PVS (9.1 kW), BESS (12 kW), and the utility grid (10 kW). Any discrepancies between generated and consumed power are accounted for as system losses. Upon $t = 2$ s, the WGS is introduced, injecting 11.8 kW into the system. Consequently, the grid transitions from supplying power to absorbing 1 kW, effectively becoming a sinking power source. At $t = 3.5$ s, the grid is fully disconnected, reducing its power contribution to zero. Despite this, the load power remains relatively constant, and the GFM inverter dynamically adjusts the battery power output to maintain system stability.

Figure 9 further demonstrates the effectiveness of the proposed GFM and GFL inverter control strategies under various operating conditions. As depicted in Fig. 9(a), the actual power coefficient closely follows its reference value,

ensuring optimal performance. Fig. 9(b) highlights the stability of the DC-link voltage, which remains consistently regulated across all scenarios. Further, Fig. 9(c) shows the grid current during connection and outage. It can be seen that the load is continuously supplied with its requirements, as confirmed from Figs. 9(d) and (e), where Fig. 9(e) presents the load voltage and current waveforms, further validating the system's reliability and effective power management. Lastly, the actual values of both GFM and GFL inverters align with their respective reference values, confirming the accuracy of the implemented control strategies, as confirmed from Figs. 9 (f)-(i).

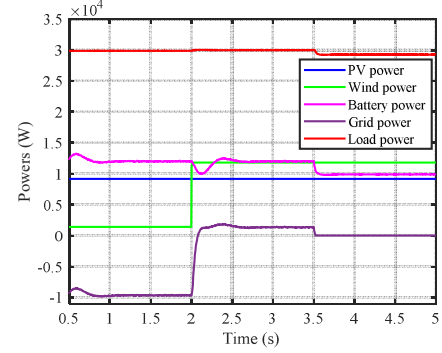


Fig. 8. All generated and sinking powers.

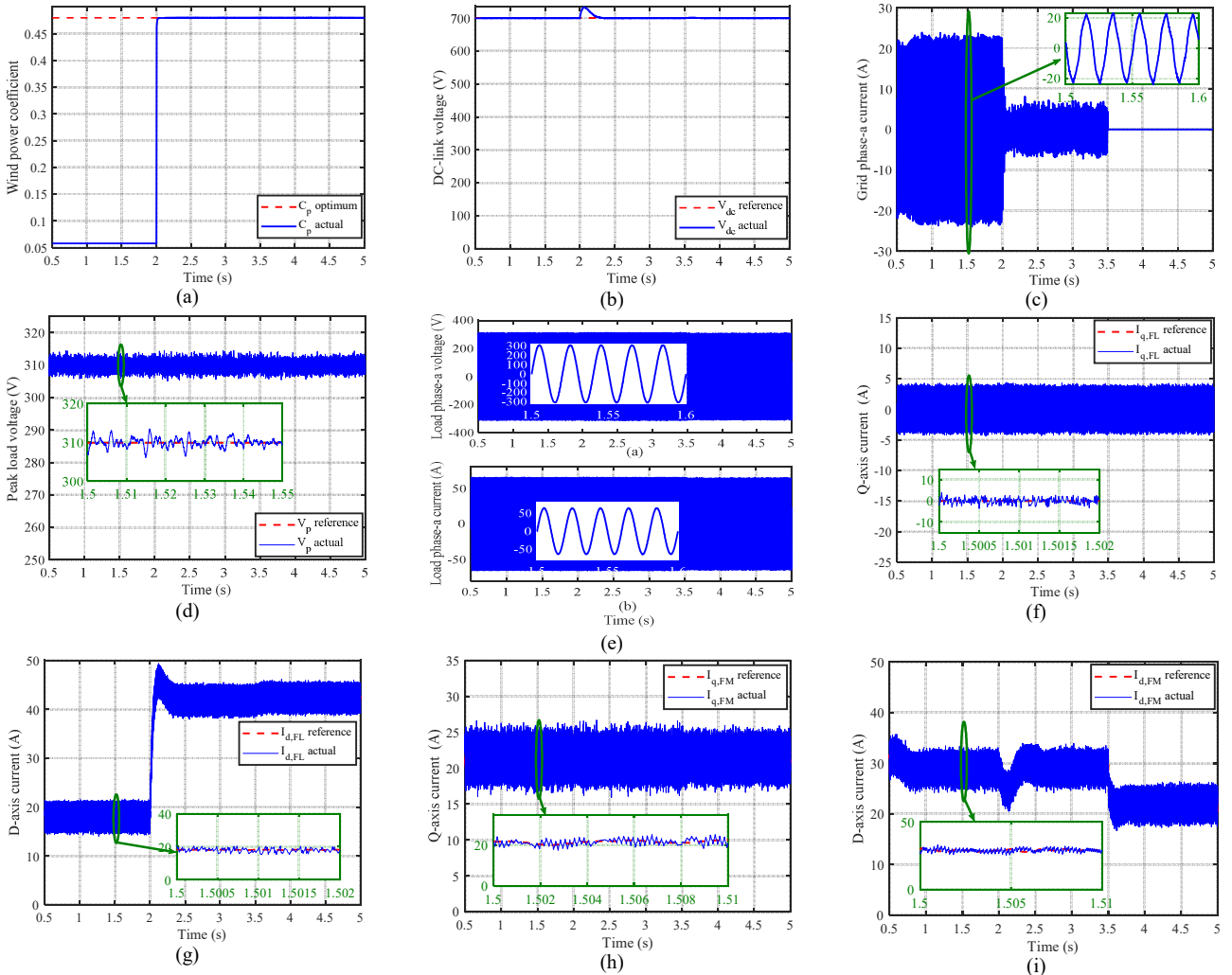


Fig. 9. Effectiveness of the GFM and GFL inverter control for the overall system performance under different circumstances. (a) Wind power coefficient, (b) Response of DC-link voltage, (c) Grid phase-a current, (d) Peak load voltage, (e) D-axis GFL inverter current, (f) Q-axis GFL inverter current, (g) single phase load voltage and current, (h) Q-axis GFM inverter current, and (i) D-axis GFM inverter current.

V. CONCLUSION

This study presented a comprehensive analysis of a GFM inverter for a BESS to enhance voltage and frequency stability in an AC microgrid. The proposed system effectively integrated RESs, including a WGS and a PVS, ensuring stable operation under varying conditions. By employing droop control, the GFM inverter maintained system stability, particularly during grid outages, while the GFL inverter ensured proper synchronization with the utility grid through vector control and a PLL. Simulation results demonstrated the effectiveness of the proposed control strategy in maintaining steady voltage and frequency, ensuring rapid response to grid disturbances, and enabling seamless transitions between grid-connected and islanded modes. The system's ability to adapt to sudden changes in power generation and load demand, while preserving a stable DC-link voltage, validates its robustness for contemporary AC microgrids. The findings highlight the significance of incorporating advanced inverter control strategies in microgrid applications, particularly in scenarios involving high penetration of RESs. Future research will focus on further optimizing control methodologies, exploring real-time hardware implementations, and investigating additional scenarios to enhance system reliability and resilience.

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